

The empirical recurrence for OEIS A236706, recovered from its tail

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Abstract

OEIS A236706 records “Empirical recurrence of order 59” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 442$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 2$, so the recurrence is proved for every $n \geq 61$. Its last coefficient is $c_{59} = -11776 \neq 0$, so no further tail satisfies anything shorter; any order-59 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A236706 is “Number of $(n+1) \times (2+1)$ 0..2 arrays colored with the maximum plus the upper median minus the lower median of every 2×2 subblock.”. It has offset 1 and begins

442, 3492, 27406, 216340, 1707838, 13499232, 106577620, 841345444, ...

The entry states:

Empirical recurrence of order 59 (see link above)

As of the “Last modified” line on the live entry (Jul 23 2025 09:27:52, revision 7) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 442$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 442$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 59: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 2$ is the first whose minimal order is exactly the stated 59.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 884$ exact terms from $n = 2$ on, it returns order 59 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{59} c_i a(n-i),$$

c_1	0	c_{16}	−521167333	c_{31}	−35898470721	c_{46}	1207899103
c_2	4	c_{17}	−261452423	c_{32}	−42203612554	c_{47}	2179512903
c_3	137	c_{18}	1212012030	c_{33}	−14695560552	c_{48}	810070054
c_4	1746	c_{19}	1143162897	c_{34}	60042757624	c_{49}	401408376
c_5	5202	c_{20}	1569652651	c_{35}	−1033074804	c_{50}	−1241957802
c_6	15874	c_{21}	1801345650	c_{36}	51836476826	c_{51}	120183296
c_7	5723	c_{22}	−6975559973	c_{37}	48550215745	c_{52}	−74378220
c_8	−303874	c_{23}	−6867102883	c_{38}	−78300321890	c_{53}	−171735472
c_9	−913347	c_{24}	−2679818238	c_{39}	50917148413	c_{54}	−20099360
c_{10}	−1785263	c_{25}	−6137192285	c_{40}	−29581638784	c_{55}	5363616
c_{11}	−367582	c_{26}	19305663398	c_{41}	−24856700786	c_{56}	5137920
c_{12}	22715911	c_{27}	24745056020	c_{42}	36690010768	c_{57}	−698112
c_{13}	52513638	c_{28}	12398294946	c_{43}	−27736957012	c_{58}	29952
c_{14}	6715396	c_{29}	7299775838	c_{44}	10782834711	c_{59}	−11776
c_{15}	−159635278	c_{30}	−36353860314	c_{45}	−6350531117		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 61$, and it is the recurrence OEIS A236706 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 2$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{59} - c_1 x^{58} - \dots - c_{59}$. Its constant term is $-c_{59} = 11776$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 59 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 59, so $\deg q = 59$, and it is monic. It holds from some index t on. From $\max(t, 2)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 59, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 17 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry's English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 59, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry's.

Third, the recovered recurrence was evaluated directly on the entry's own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is -11776 .

Remark 4. The conjecture's own statement was never read. The entry pins it down to the family of order-59 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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